

SEAT No. _____

[55]

SARDAR PATEL UNIVERSITY No. of Printed Pages : 02

BCA Semester-6 Examination, 2021

US06CBCA23 || Data Communication and Computer Networks

Thursday, 09th September 2021

Time : 02:00 TO 04:00 P.M.

09/09/2021

Marks: 70

Q-1 Select the appropriate option for the following questions:

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- 1) The data elements sent from one device to another per second is known as _____.
A. Bandwidth
B. Modulation rate
C. Data rate
D. None of these
- 2) _____ is to make all programs, equipments & specially data available to anyone on a network without regard to the physical location of the resource & user.
A. High reliability
B. Saving money
C. Resource-sharing
D. Scalability
- 3) Which of the following physical connection is the fastest?
A. Twisted pair
B. Fiber-optic cable
C. Coaxial cable
D. Radio waves.
- 4) For long distance data transmission, what is the preferable mode of communication?
A. Serial transmission
B. Simplex
C. Parallel transmission
D. Half duplex
- 5) Routing is responsibility of _____.
A. Data link layer
B. Network layer
C. Transport layer
D. Physical layer
- 6) The signal from a satellite is normally aimed at a specific area called _____.
A. Channel
B. Tower
C. Footprint
D. Orbit
- 7) Which of the following CSMA has highest chances of collision?
A. 1-persistent
B. non-persistent
C. p-persistent
D. All have equal chances
- 8) A switch performs filtering on the basis of _____ address.
A. MAC
B. port
C. logical
D. postal
- 9) When a beam of light travels through media of two different densities, if the angle of incidence is greater than the critical angle, _____ occurs.
A. Reflection
B. Incidence
C. Refraction
D. Criticism
- 10) The connection of two or more networks is called an _____.
A. computer network
B. internetwork
C. internet
D. local Area Network

①

(P.T.O.)

Q-2 **Tick (TRUE or FALSE) and Fill in the blanks.**

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- 1) Subnet consists of two distinct components: transmission lines & switching elements. [TRUE/FALSE]
- 2) In circuit switching, each switch stores the whole message & forwards it to the next switch. [TRUE/FALSE]
- 3) OSI stands for Open Standard Interconnection. [TRUE/FALSE]
- 4) A repeater operates at Network layer. [TRUE/FALSE]
- 5) DQDB stands for _____.
- 6) Transmission media are usually categorized as _____.
- 7) A _____ is a set of rules that govern data communications.
- 8) CSMA stand for _____.

Q-3 **Give short answers of the following questions: (ANY TEN)**

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- 1) List the advantages of computer networks.
- 2) Define terms: Topology, Server
- 3) Explain DQDB in brief.
- 4) Explain parallel transmission in brief.
- 5) Differentiate between TDM and FDM.
- 6) Write the difference between circuit switching and packet switching.
- 7) Explain orbit.
- 8) List categories of satellite.
- 9) Define: Protocol Stack and Network architecture.
- 10) Explain in brief: Broadcast network.
- 11) What is the difference between bridge and repeater?
- 12) What is the function of Modem?

Q-4 **Give Long answers of the following questions: (ANY FOUR)**

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- 1) How can we classify computer Network? Explain.
- 2) Explain LAN in detail.
- 3) Explain fiber optic in detail.
- 4) What do you mean by Multiplexing? Explain FDM in detail.
- 5) List all layers of OSI model. Explain any three layers.
- 6) What is communication satellite? What are the types of it? Explain geosynchronous satellite.
- 7) Explain star topology and mesh topology with their advantages & disadvantages.
- 8) Explain working of Repeater & Router.

